

ROAD TRAFFIC ACCIDENT HOTSPOT DETECTION AND SAFETY ANALYSIS IN NIGERIA



A CAPSTONE PROJECT REPORT

BY

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JUNE, 2025.

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EXECUTIVE SUMMARY

This report presents a comprehensive analysis of road traffic accidents in Nigeria using real-world data from the Federal Road Safety Corps (FRSC) covering Q1 2023 to Q3 2024. The project aimed to identify accident hotspots, analyze accident severity and causes, and generate data-driven recommendations to improve road safety. Using Microsoft Excel, SQL Server, and Power BI, we explored accident frequency, quarterly trends, top causative factors, and casualty impacts across Nigerian states. Key findings revealed the Federal Capital Territory (FCT), Ogun, and Kaduna as high-risk regions, with speed violation accounting for over 40% of accident causes. This project provides a valuable foundation for policymakers and transportation authorities to implement targeted interventions and reduce accident-related fatalities.

ACKNOWLEDGMENT

We, Group 21 of the TechCrush Data Analytics Track, wish to express our heartfelt gratitude to God Almighty for granting us the grace, wisdom, and perseverance to successfully complete this capstone project. We deeply appreciate the unwavering support of our Student Success Executive (SSE), Ms. Mary Oladokun, and our mentors, Ms. Grace Emeruwa and Mr. Olaniyi Eluwole, whose expert guidance and encouragement were invaluable throughout this journey. We also extend sincere thanks to the Federal Road Safety Corps (FRSC) for providing access to public datasets that enabled this real-world analysis.

We recognize the collective efforts and dedication of our team members, each of whom contributed to different phases of the project. Below is an acknowledgment of their contributions, organized by their primary roles within the workflow:

Data Cleaning, Transformation, and Preparation

- Ogechi Adiole – Data cleaning, standardization, and transformation.
- Muhammed Rufai – Excel-based data cleaning, Power Pivot, Pivot Table, and Dashboard setup.
- Toba Adeshina – Assisted in data validation and transformation tasks.

Data Extraction, Querying, and Analysis

- Ewoma Zinno – SQL development and project report writing
- Ahortor Godslove – SQL querying and ETL (Extract, Transform, Load) processes.
- Emmanuel Obiesan – Assisted in query testing and dataset troubleshooting.
- Obademi Grace: Assisted in data extraction.

Domain Knowledge & Interpretation

- Precious Okit – Defined acronyms from accident causative factor tables (e.g., SPV, TBT).
- Akinwumi Hezekiah Akintunde – Assisted in analyzing accident trends.
- Famure Olatomiwa – Supported analysis and interpretation of accident causes.
- Oyeshide Victoria – Helped with preliminary topic brainstorming and research inputs.
- Owo Moses – Conducted road accidents research.

Visualization, Presentation & Reporting

- Ikuesiri Sonia Edebiri – Project proposal drafting and PowerPoint presentation design.
- Terungwa Emmanuel – Power BI visualizations final report input.
- Alih Emmanuel O. – Power BI dashboard development and final report writing.

INTRODUCTION

Road traffic accidents (RTAs) remain a pressing public health and safety challenge in Nigeria. According to the Federal Road Safety Corps (FRSC), thousands of Nigerians are injured or killed annually due to preventable road crashes. Contributing factors include excessive speeding, poor road infrastructure, vehicle conditions, driver behavior, and ineffective emergency response systems. Despite ongoing awareness campaigns, the lack of **data-driven interventions** limits the effectiveness of road safety policies and planning.

This project leverages real-world accident data from the FRSC to analyze the **severity**, **frequency**, and **causative factors** of road accidents in Nigeria. Using business intelligence tools and structured data workflows, this project identifies key trends, accident hotspots, and high-risk behaviors, with the ultimate goal of providing actionable safety recommendations to reduce road fatalities and improve transportation safety.

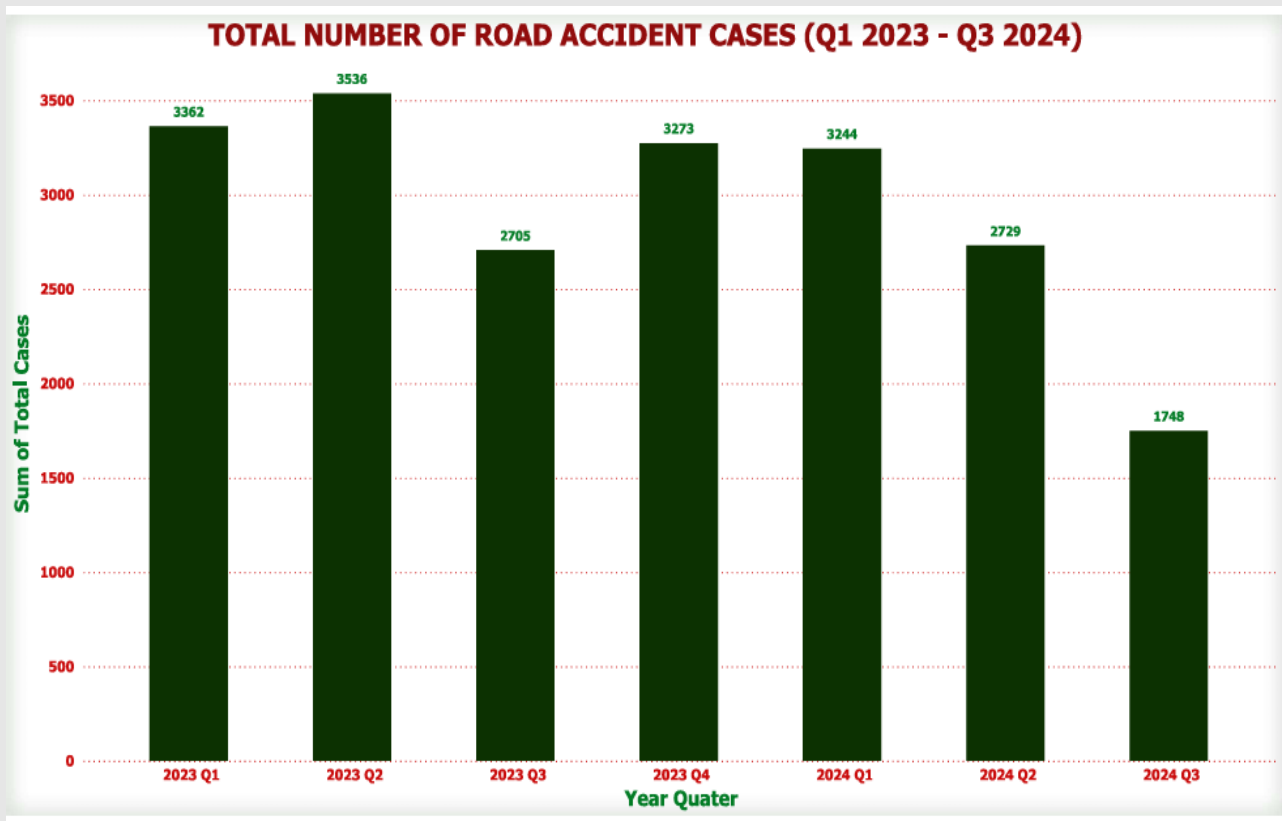


Figure 1: Total Number of Accident Cases

PROJECT OBJECTIVES

The primary objectives of this capstone project are to:

1. Analyze the frequency and severity of road accidents per state in Nigeria.
2. Identify patterns and trends based on accident severity (fatal, serious, minor).
3. Determine the most common causes of road accidents nationally and per state.
4. Visualize the total casualties and people involved using interactive dashboards.
5. Generate actionable safety recommendations based on insights from the analysis.

DATA OVERVIEW

Data Source: National Bureau of Statistics (NBS) & Federal Road Safety Corps (FRSC), Q1 2023 – Q3 2024.

Datasets Used:

1. Road Accident Severity Data
 - Fields: STATE, STATEID, YEAR, QUARTER, FATAL, SERIOUS, MINOR, TOTAL_CASES, NUMBER_INJURED, NUMBER_KILLED, TOTAL_CASUALTY, PEOPLE_INVOLVED.
2. Road Accident Causative Factors Data
 - Fields: STATE, STATEID, YEAR, QUARTER, SPV (Speed Violation), BFL (Brake Failure), MDV (Mechanical Defect), OVL (Overloading), DGD (Dangerous Driving), and others.

Tools Used:

Tool	Purpose
Microsoft Excel	Initial data cleaning and formatting
SQL Server	Data querying, aggregation, and modeling
Power BI	Dashboard development and visualization

Table 1: Tools Used

Reference Table for Road Accident Causes

ACRONYM	FULL MEANING
SPV	Speed Violation
UPD	Use of Phone while Driving
TBT	Tyre Burst
MDV	Mechanically Deficient Vehicle
BFL	Brake Failure
OVL	Overloading
DOT	Dangerous Overtaking
WOT	Wrongful Overtaking
DGD	Dangerous Driving
BRD	Bad Road
RTV	Route Violation
OBS	Road Obstruction Violation
SOS	Sleeping on Steering
DAD	Drinking Under Alcohol/Drug Influence
PWR	Poor Weather
FTQ	Fatigue
SLV	Sign Light Violation
OTH	Others

Table 2: Reference Table for Road Accident Causes

ANALYSIS PROCESS

1. Cleaned and standardized datasets using Excel and added unique STATEID for joins.
2. Imported the datasets into SQL Server and wrote optimized queries for:
 - Frequency and severity summaries per state
 - Quarterly trends
 - Ranking of top causes per state and nationwide
 - Casualty and involvement aggregation
3. Used Power BI to build dynamic visuals and dashboards for stakeholder presentation.
4. Extracted key insights and synthesized actionable recommendations.

KEY FINDINGS AND INSIGHTS

Objective 1: Accident Frequency & Severity

- ✓ FCT recorded the highest accident frequency across both years.
- ✓ Serious accidents accounted for over 50% of total reported cases.
- ✓ Other high-risk states include Ogun, Lagos, Kaduna, and Rivers.

Objective 2: Patterns and Trends

- ✓ Q2 recorded the most accidents in both years.
- ✓ Quarterly peaks suggest seasonal travel behaviors and weather conditions.
- ✓ Minor accidents are increasing slightly, suggesting a shift in accident type severity.

Objective 3: Top Causes of Accidents

- ✓ Speed Violation (SPV) was the leading cause (40–45% of all cases).

- ✓ Other major causes include Brake Failure, Dangerous Driving, and Overloading.
- ✓ Trends show that behavioral causes are far more common than infrastructure-related ones.

Objective 4: Casualties and People Involved

- ✓ Some states like Kano and Bauchi had high casualty-to-case ratios.
- ✓ Over 12,000 people were involved in the recorded crashes.
- ✓ Fatality rates were highest where emergency response and road conditions were poor

DASHBOARDS

Dashboard 1: Road Accident Trends and Severity Overview

Visual	Description
Bar Chart – Accidents per State	Shows total accident count per state.
Line Chart – Quarterly Trends	Visualizes fatal, serious, and minor cases over time.
Stacked Column – Severity Over Time	Combines all accident types per quarter for trend comparison.
Nigeria Map (Shape Map)	Darker colors show higher accident states. FCT, Ogun, and Kaduna stand out.

Table 3: Dashboard 1 Visual Description

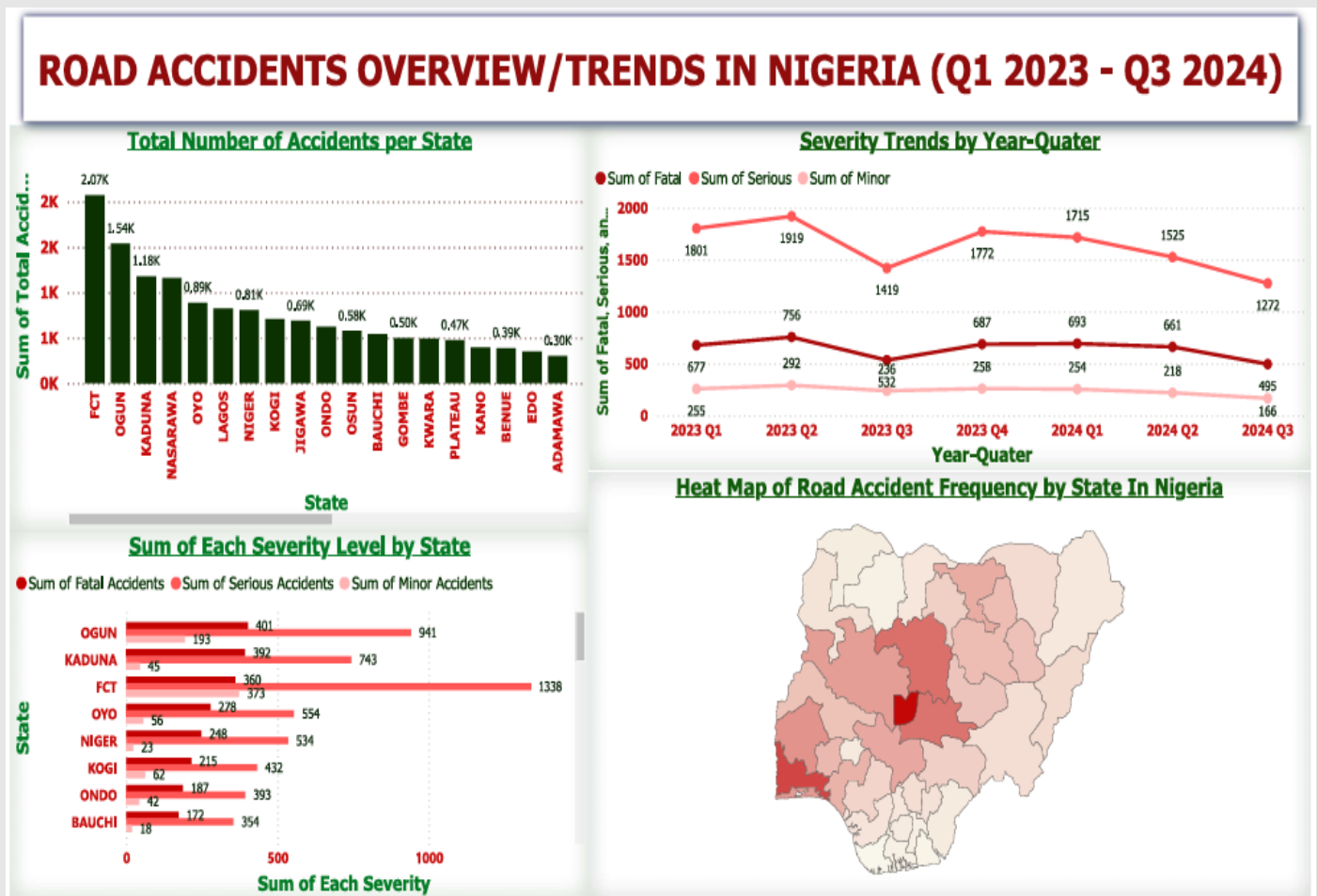


Figure 2: Dashboard 1

Dashboard 2: Casualties and People Involved

Visual	Description
Bar Chart – Casualties per State	Total number of people injured or killed per state.
Card Visuals – National Totals	Shows overall number of injuries, deaths, and people involved.
Comparative Bar Chart	Displays people killed vs injured per state for context.

Table 4: Dashboard 2 Visual Description

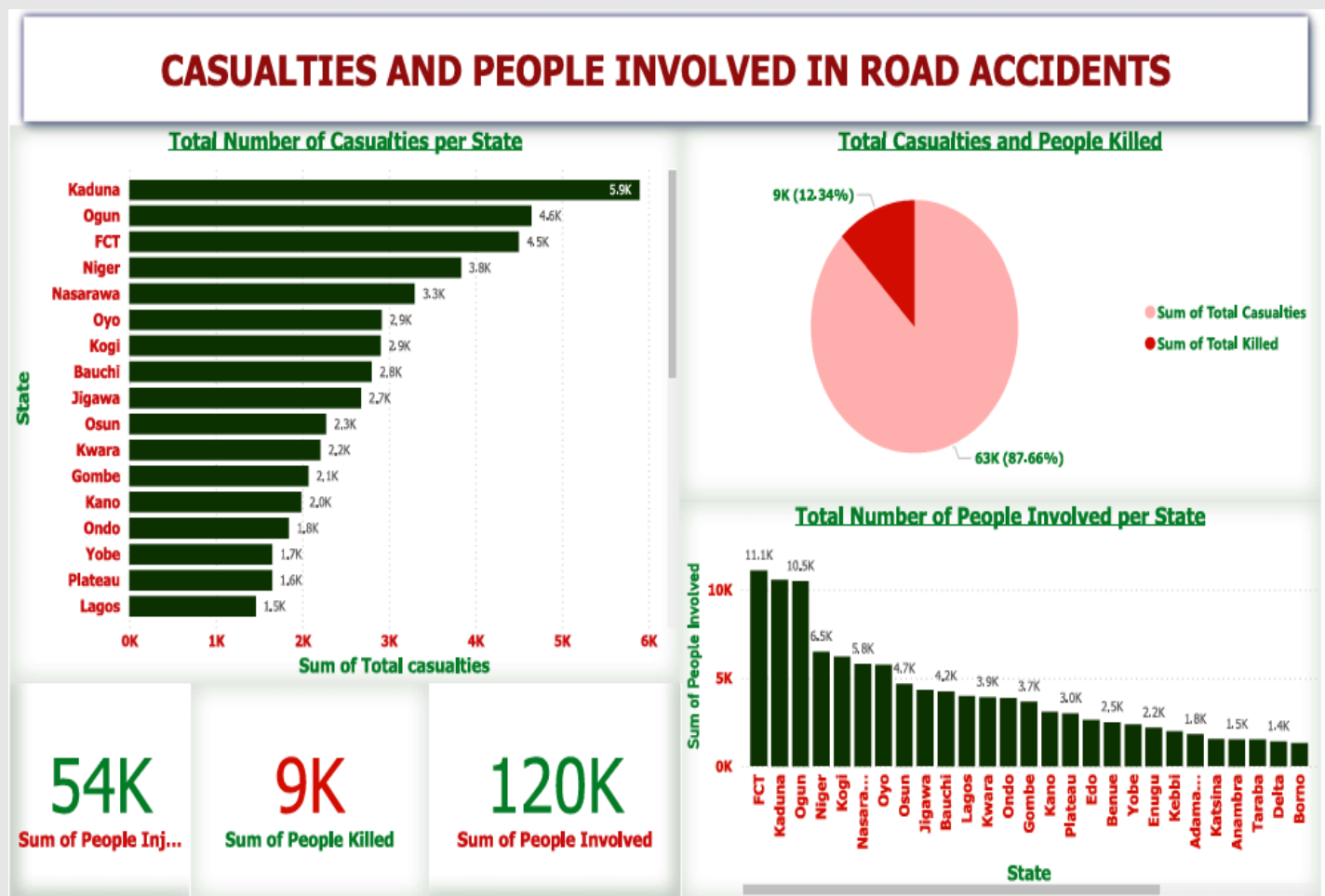


Figure 3: Dashboard 2

Dashboard 3: Top Causes of Accidents

Visual	Description
Bar Chart – Nationwide Causes	Total count of each causative factor.
Line/Bar – Trends by Cause over Time	Identifies changes in accident causes quarterly.
Matrix – Top 5 Causes per State & Quarter	Lists dominant causes of accidents in each state over time.
Slicers	Enable filtering by YEAR, QUARTER, and STATE for detailed exploration.

Table 5: Dashboard 3 Visual Description

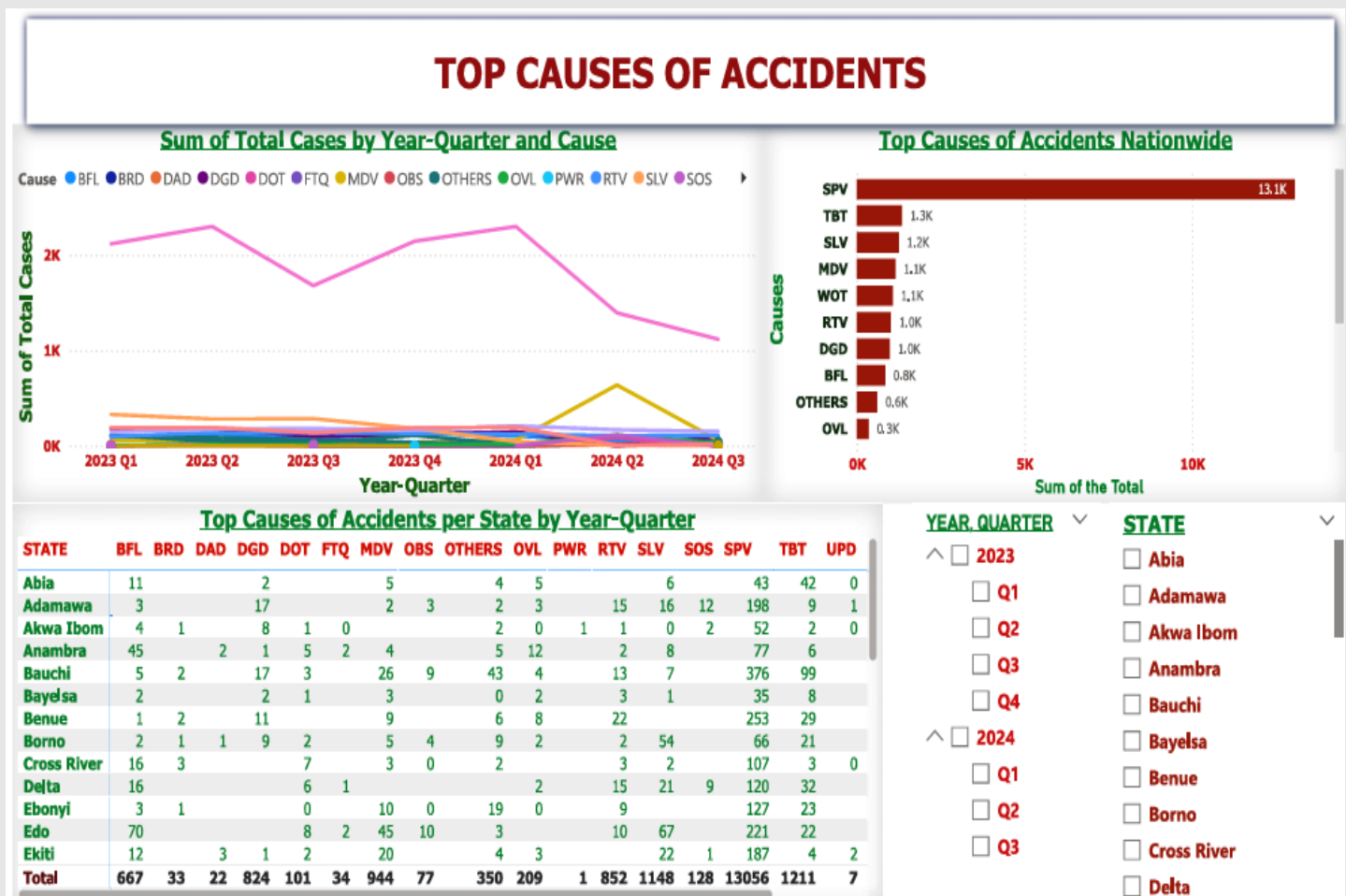
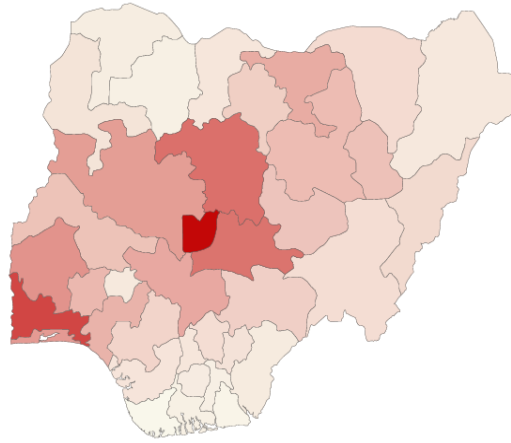


Figure 4: Dashboard 3

HEAT MAP OF ROAD ACCIDENT FREQUENCY BY STATE

This map shows the total road accidents per state in Nigeria. Darker shades represent higher accident counts, with FCT, Ogun, and Kaduna emerging as high-risk areas between Q1 2023 and Q3 2024. It also highlights key hotspots for targeted safety actions.



TOP CAUSES OF ACCIDENTS PER STATE BY YEAR-QUARTER

STATE	BFL	BRD	DAD	DGD	DOT	FTQ	MDV	OBS	OTHERS	OVL	PWR	RTV	SLV	SOS	SPV	TBT	UPD	WOT	Total
Abia	11			2			5		4	5			6		43	42	0	5	123
Adamawa	3			17			2	3		2	3		15	16	12	198	9	1	29
Akwa Ibom	4	1		8	1	0			2	0	1	1	0	2	52	2	0	13	87
Anambra	45		2	1	5	2	4		5	12		2	8		77	6		30	199
Bauchi	5	2		17	3		26	9	43	4		13	7		376	99		39	643
Bayelsa	2			2	1		3		0	2		3	1		35	8		7	64
Benue	1	2		11			9		6	8		22			253	29		25	366
Borno	2	1	1	9	2		5	4	9	2		2	54		66	21		20	198
Cross River	16	3			7		3	0	2			3	2		107	3	0	19	165
Delta	16				6	1				2		15	21	9	120	32		17	239
Ebonyi	3	1			0		10	0	19	0		9			127	23		15	207
Edo	70				8	2	45	10	3			10	67		221	22		10	468
Ekiti	12		3	1	2		20		4	3			22	1	187	4	2	2	263
Enugu	68			11	14		7	1	11	4		7	4		183	7		12	329
FCT				95			84					141	523	13	1691	98			2645
Gombe	22	2		53			21					11	22		276	29	2	21	459
Imo	20	1	3	9	1		1		4	3		0			84	8		8	142
Jigawa	3			8	27		30	45	14	33		8			786	73		23	1050
Kaduna				85	7	18			12	6		84		13	673	88		60	1046
Kano	3			30			26		6	3		20	40	4	276	7		27	442
Katsina		2		3	0		8	2	10	6		3	10	2	162	20		11	239
Kebbi	0			13	1		13	0	2	4		4	1		164	1	0	2	205
Kogi	46				7		24		10	21		14	57	15	491	76		58	819
Kwara	20			12			38		4	8		8	57		388	33		84	652
Lagos	62		3	15	9		66		22	11		18			470			33	709
Nasarawa				138			39		51			89	143	15	799	33		19	1326
Niger	7	8		22		7	39		10	12		14	11		703	26		59	918
Ogun	57			9			135		43			119	13	24	1703	174		73	2350
Ondo	12			55		4	51			8		44			479	38		55	746
Osun	13				4		43		8	7		66		13	428	62		29	673
Oyo	53			31	8		81		12	14		36			657	56		44	992
Plateau	76			37	11		53	2	12			12			224	6		28	461
Rivers	10		2	4	1		5	0	1			12		2	87	7		3	134
Sokoto	5			26	1		19		3	6		11	2		119	3	1	12	208
Taraba		6		59			7		5			19	48		86	17		9	256
Yobe				21			17	1	16	13		13	4		183	49		21	338
Zamfara	0	4		1	2		5		0	4		4	9	3	82		1	11	126
Total	667	33	22	824	101	34	944	77	350	209	1	852	1148	128	13056	1211	7	933	20597

TOP 5 ACCIDENT HOTSPOTS (Q1 2023 - Q3 2024)

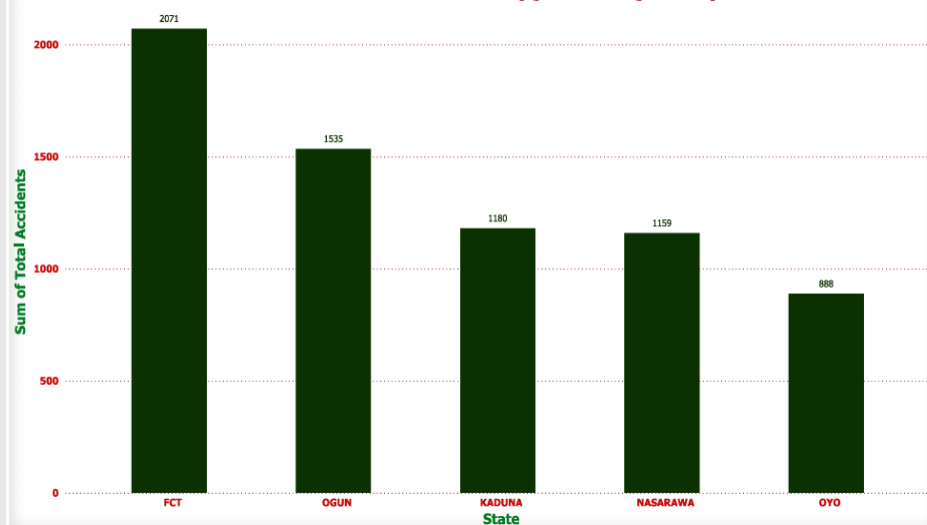


Figure 5: Other Visuals 1

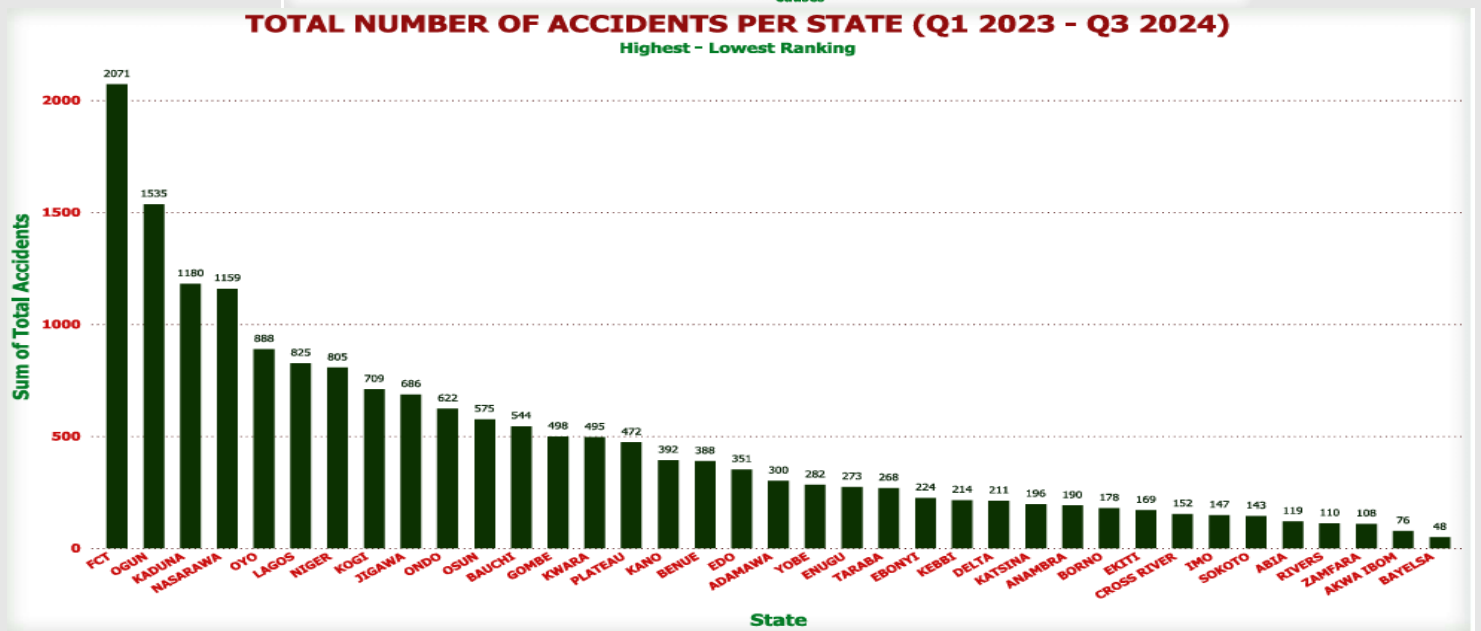
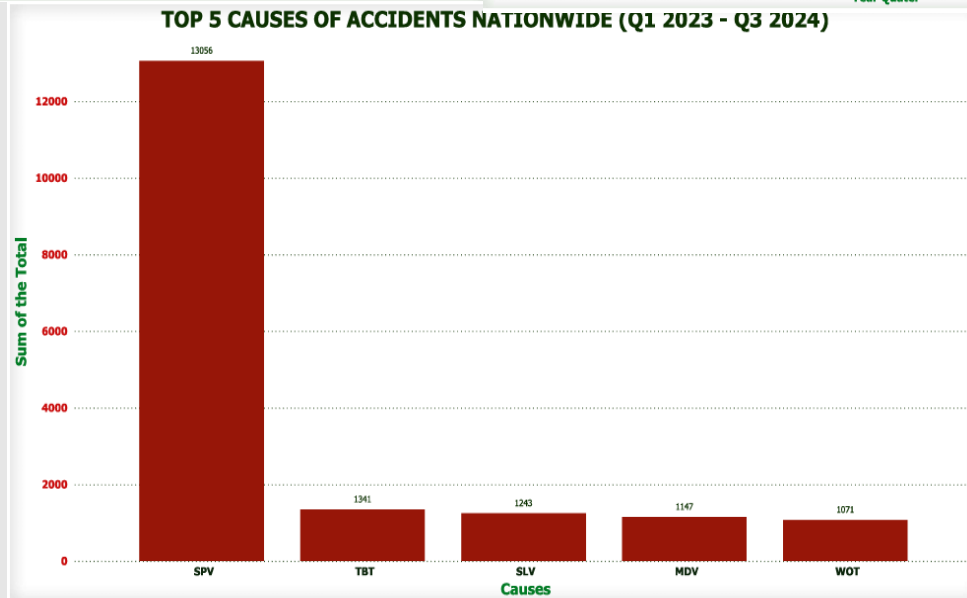
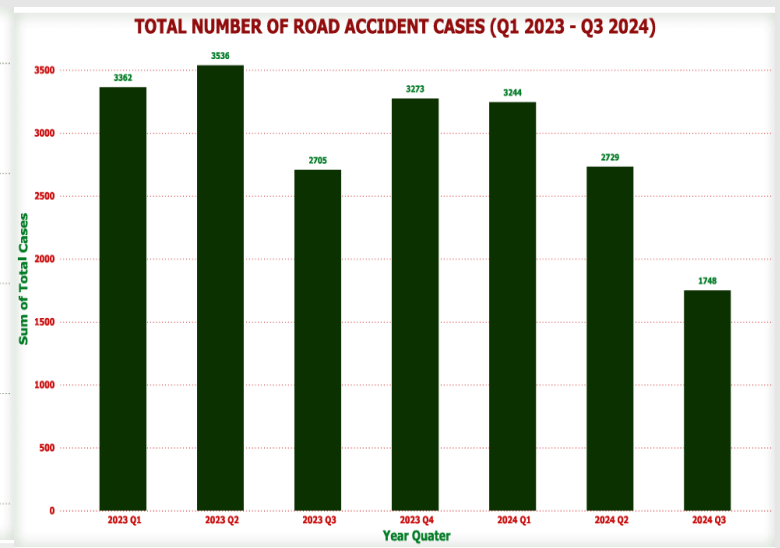
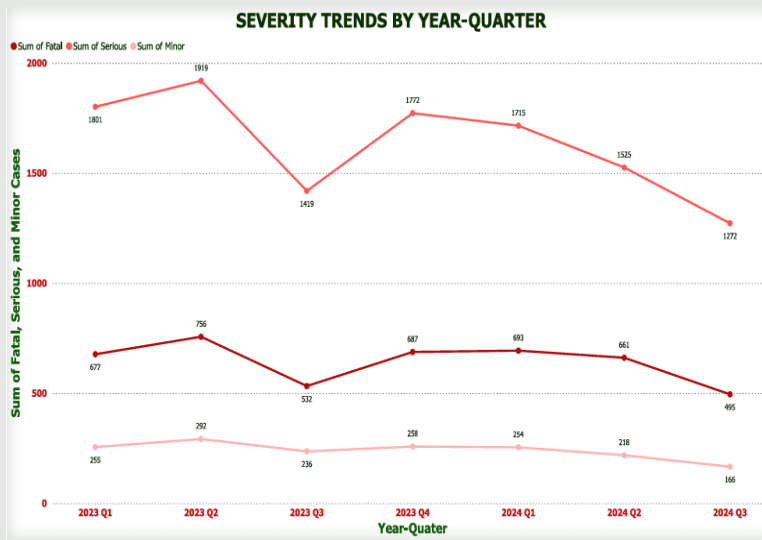


Figure 6: Other Visuals 2

RECOMMENDATIONS

Based on the data analysis and observed trends, the following are recommended:

1. Policy & Enforcement

- Install speed cameras and increase road patrols in the top 5 accident-prone states.
- Enforce rest periods for inter-state drivers to reduce fatigue-related crashes.
- Launch quarterly campaigns focusing on SPV and MDV (especially before Q2).

2. Infrastructure Improvements

- Prioritize fixing roads in regions with high reports of brake failure and bad roads.
- Create additional bypass roads to reduce congestion and driver frustration.

3. Emergency Response

- Expand emergency response facilities in fatality-dense regions.
- Promote real-time crash reporting and response coordination with FRSC.

CONCLUSION

This capstone project has demonstrated the value of data analytics in addressing public safety concerns like road traffic accidents. By analyzing FRSC data from Q1 2023 to Q3 2024, we have identified high-risk states, dangerous accident trends, and critical causative factors.

These insights are essential for shaping effective road safety policies, allocating resources, and launching targeted interventions. With the right enforcement and infrastructure reforms, Nigeria can make measurable progress in reducing road accidents and saving lives.